

CANONICAL FORMS

- Minterm
 - Product (*AND* function)
 - Contains all variables
 - Evaluates to '1' for a specific combination

Example

$A = 0$ $\bar{A}$ B C

$B = 0$ $(0) \cdot (0) \cdot (0)$

$C = 0$

- Also called standard Product

	A	B	C	Minterm	
0	0	0	0	m_0	$\bar{A} \bar{B} \bar{C}$
1	0	0	1	m_1	$\bar{A} \bar{B} C$
2	0	1	0	m_2	$\bar{A} B \bar{C}$
3	0	1	1	m_3	$\bar{A} B C$
4	1	0	0	m_4	$A \bar{B} \bar{C}$
5	1	0	1	m_5	$A \bar{B} C$
6	1	1	0	m_6	$A B \bar{C}$
7	1	1	1	m_7	$A B C$

CANONICAL FORMS

- Maxterm
 - Sum (*OR* function)
 - Contains all variables
 - Evaluates to '0' for a specific combination

Example

$A = 1$ $\overline{A}$ B $\overline{C}$

$B = 1$ (1) + (1) + (1)

$C = 1$

- Also called standard sums

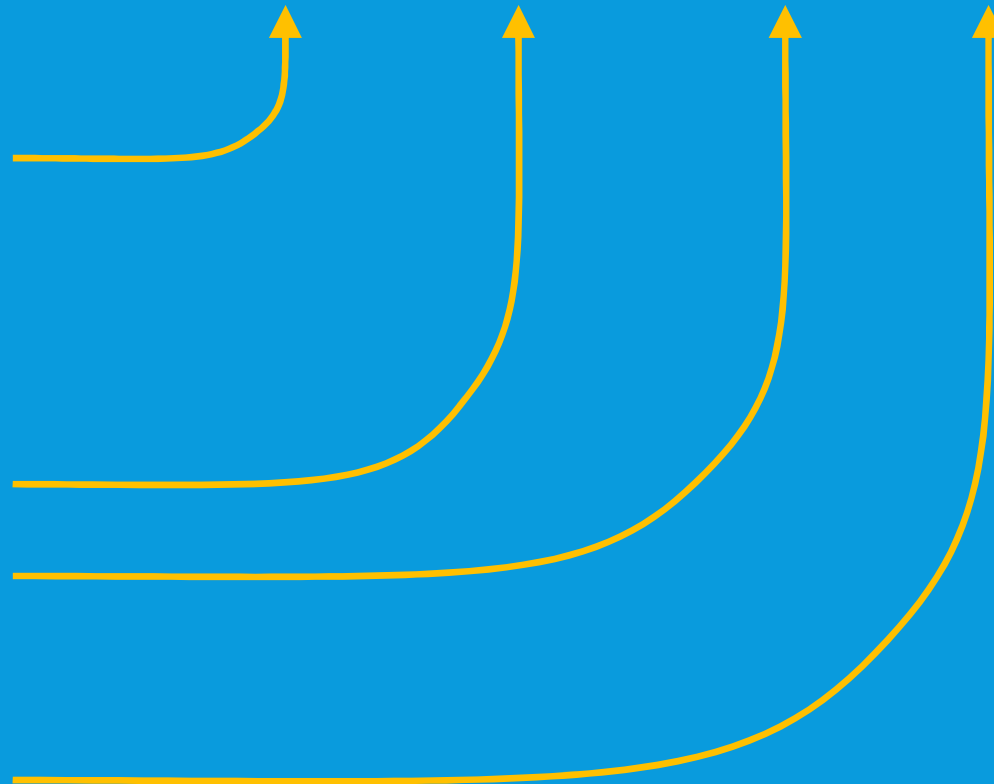
	A	B	C	Maxterm	
0	0	0	0	M_0	$A + B + C$
1	0	0	1	M_1	$A + B + \overline{C}$
2	0	1	0	M_2	$A + \overline{B} + C$
3	0	1	1	M_3	$A + \overline{B} + \overline{C}$
4	1	0	0	M_4	$\overline{A} + B + C$
5	1	0	1	M_5	$\overline{A} + B + \overline{C}$
6	1	1	0	M_6	$\overline{A} + \overline{B} + C$
7	1	1	1	M_7	$\overline{A} + \overline{B} + \overline{C}$

CANONICAL FORMS

Truth Table to Boolean Function

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

$$F = \overline{A}\overline{B}C + \overline{A}\overline{B}\overline{C} + \overline{A}BC + ABC$$



A Boolean function can be expressed algebraically from a given truth table by forming a minterm for each combination of the variables that produces a 1 in the function and then taking the OR of all those terms.

CANONICAL FORMS

Sum of *Minterms*

$$F = \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC$$

$$F = m_1 + m_4 + m_5 + m_7$$

$$F = \sum (1, 4, 5, 7)$$

• Product of *Maxterms*

$$\overline{F} = \overline{A}\overline{B}\overline{C} + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC$$

$$\overline{\overline{F}} = \overline{\overline{A}\overline{B}\overline{C} + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC}$$

$$F = \overline{\overline{A}\overline{B}\overline{C}} \cdot \overline{\overline{A}B\overline{C}} \cdot \overline{A\overline{B}\overline{C}} \cdot \overline{ABC}$$

$$F = (A + B + C)(A + \overline{B} + C)(A + \overline{B} + \overline{C})(\overline{A} + \overline{B} + C)$$

$$F = M_0 \quad M_2 \quad M_3 \quad M_6$$

$$F = \prod (0, 2, 3, 6)$$

$$m'_j = M_j$$

	A	B	C	F	$\overline{F}$
0				0	1
1	0	0	1	1	0
2	0	1	0	0	1
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	1	0
6	1	1	0	0	1
7	1	1	1	1	0

STANDARD FORMS

$$F = \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC$$

$$\begin{aligned} & \overline{A}\overline{B}(\overline{C} + C) \\ &= \overline{A}\overline{B}(1) \\ &= \overline{A}\overline{B} \end{aligned}$$

$$\begin{aligned} & AC(\overline{B} + B) \\ &= AC \end{aligned}$$

$$\begin{aligned} & \overline{B}C(\overline{A} + A) \\ &= \overline{B}C \end{aligned}$$

$$F = \overline{B}C(\overline{A} + A) + \overline{A}\overline{B}(\overline{C} + C) + AC(\overline{B} + B)$$

$$F = \overline{B}C + \overline{A}\overline{B} + AC$$

STANDARD FORMS

$$\overline{F} = \overline{A}\overline{B}\overline{C} + \overline{A}B\overline{C} + \overline{A}BC + A\overline{B}\overline{C}$$

$$\overline{A}B(\overline{C} + C)$$

$$B\overline{C}(\overline{A} + A)$$

$$\overline{A}\overline{C}(\overline{B} + B)$$

$$\overline{F} = \overline{A}\overline{C}(\overline{B} + B) + \overline{A}B(\overline{C} + C) + B\overline{C}(\overline{A} + A)$$

$$\overline{\overline{F}} = \overline{\overline{A}\overline{C} + \overline{A}B + B\overline{C}}$$

$$F = (A + C)(A + \overline{B})(\overline{B} + C)$$

STANDARD FORMS

- Standard form to Canonical form

$$F = A + B'C$$

$$= A(B+B') + B'C(A+A')$$

$$= AB + AB' + AB'C + A'B'C$$

$$= AB(C+C') + AB'(C+C') + AB'C + A'B'C$$

$$= ABC + ABC' + AB'C + AB'C' + AB'C + A'B'C$$

$$= A'B'C + AB'C' + AB'C + ABC' + ABC$$

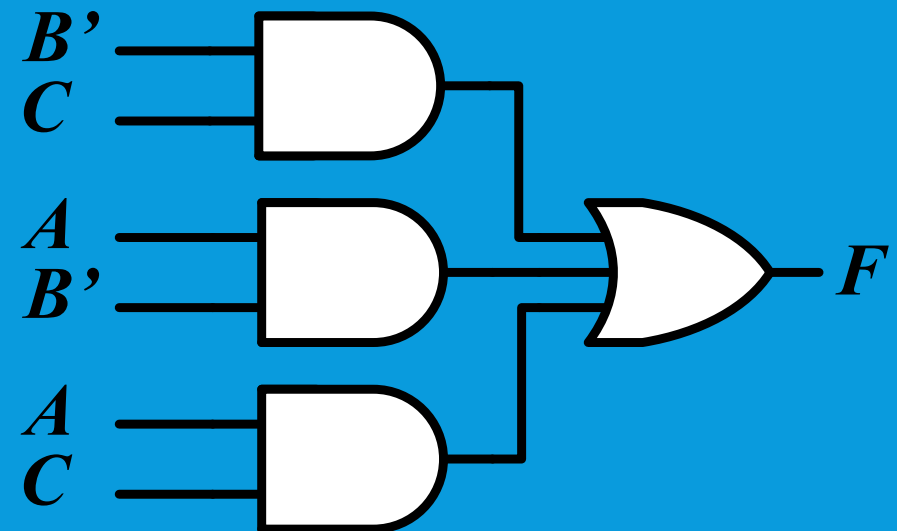
$$= m_1 + m_4 + m_5 + m_6 + m_7$$

$$F(A,B,C) = \Sigma(1,4,5,6,7)$$

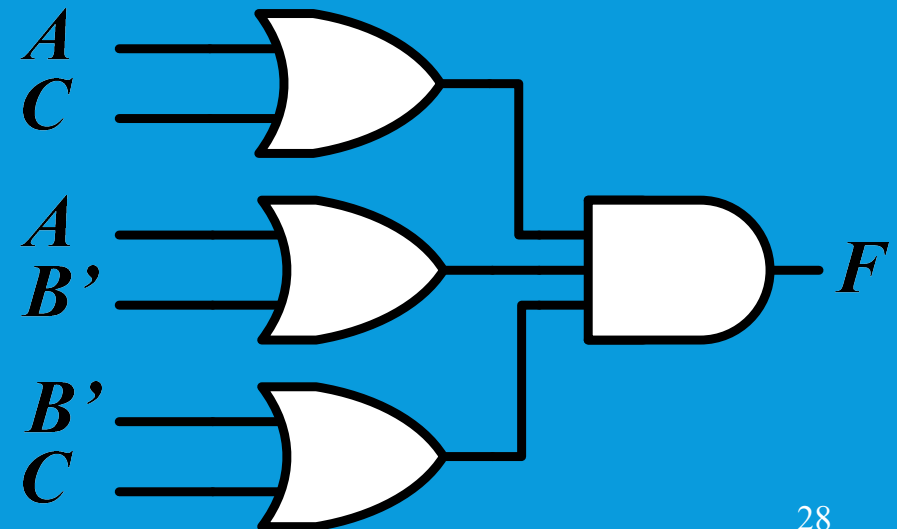
OR you can use the truth table for F

TWO LEVEL IMPLEMENTATION

- **Sum of Products (SOP)**



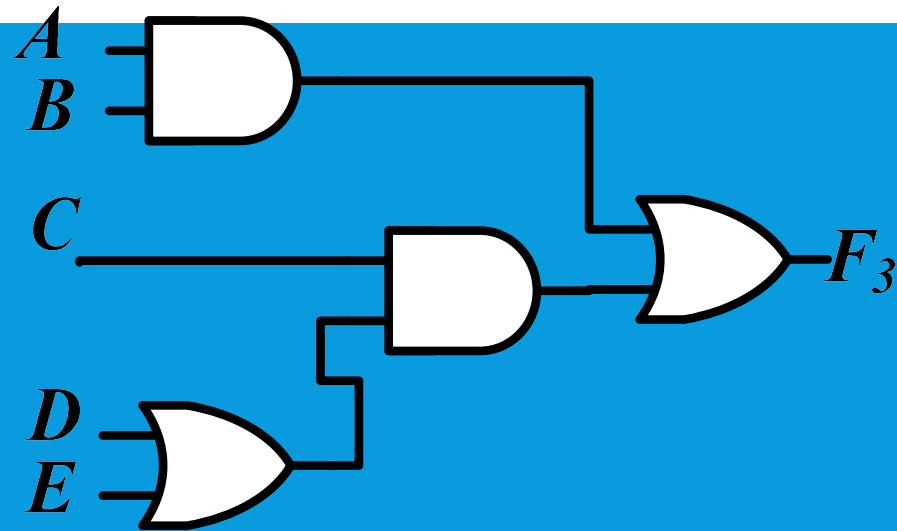
- **Product of Sums (POS)**



THREE AND TWO LEVELS IMPLEMENTATION

- Three Levels

$$F_3 = AB + C(D + E)$$



- Two Levels

$$F_3 = AB + CD + CE$$

